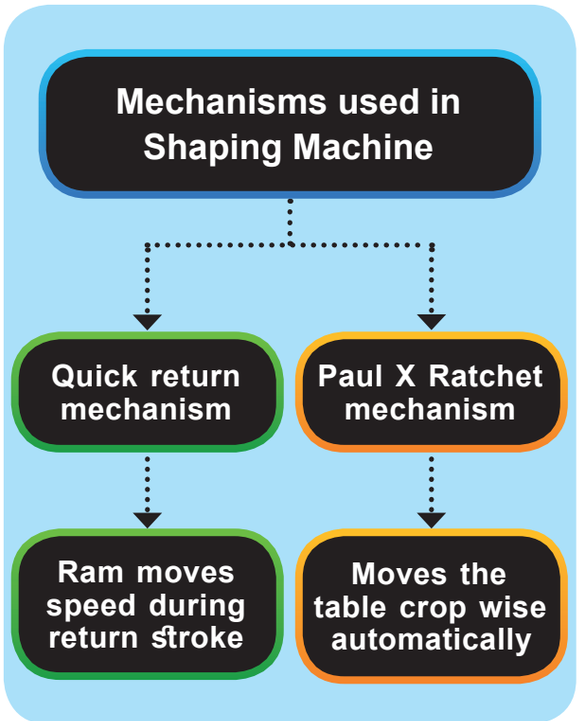


3.9 Pawl and Ratchet mechanism

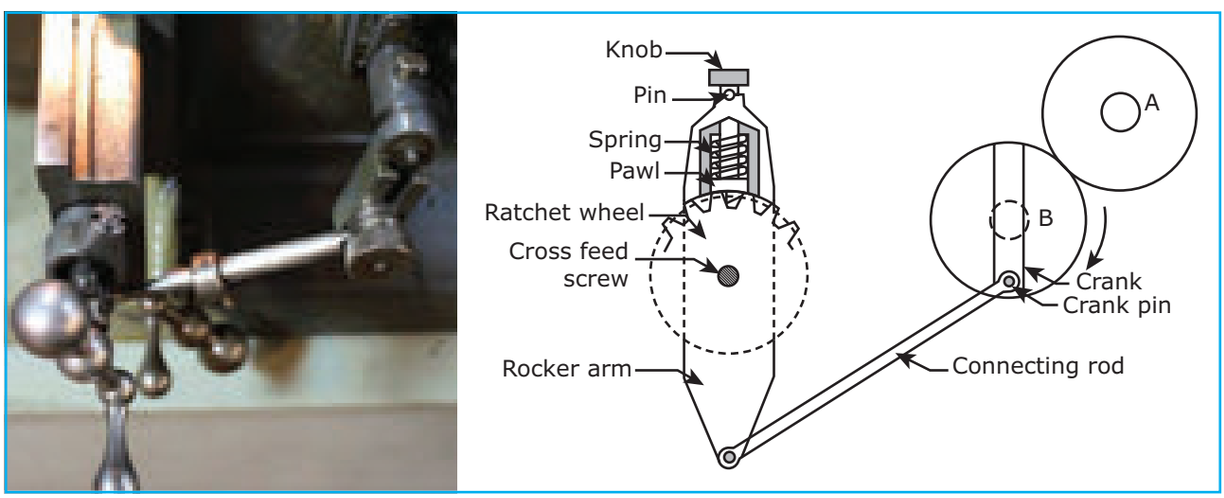
The table of a shaping machine travels in a cross direction when the crossfeed screw is rotated. The cross feed screw is attached to the ratchet wheel. A spring 'pawl' is positioned to be placed between the teeth of the ratchet wheel. The pawl is housed within a frame known as rocker arm. The bull gear placed inside the column of the shaping machine drives the gear B through the gear A.

There is a diametric slot provided on the face of the gear B. A crank pin is attached to a slider placed in the slot. The bottom of the rocker arm and crank pin are connected by a connecting rod. The rotation of the gear B makes the crank pin to rotate. This movement makes the rocker arm to rock about the centre of the ratchet wheel. The pawl makes the ratchet to rotate by a small amount in one direction only. As the cross feed screw is attached to the ratchet wheel, the rotation of the ratchet wheel will make the table to move in a cross direction.

If the direction of the table feed is to be reversed, the pawl is turned about 180^0 from its position. The ratchet wheel and the crossfeed screw will rotate in the opposite direction resulting in the table movement in the opposite direction.

Pawl & Ratchet mechanism is shown in Figure

When power feed is not necessary for the table, the pawl is disengaged from the ratchet wheel.



Pawl and Ratchet Mechanism

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